Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 270 (2025) 4726–4735

Procedia Computer Science

www.elsevier.com/locate/procedia

29th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2025)

Navigating Responsible AI: A Systematic Review of Governance Mechanisms and Future Co-Governance Scenarios

Imen Ferjani[a], Fatimah Abdulraheem Alzahrani[a], Noureddine Mohamed Bouzir[b]

[a] *Computer Department, Deanship of Preparatory Year and Supporting Studies,Imam Abdulrahman Bin Faisal University,Dammam 31441, Kingdom of Saudi Arabia*

[b] *Ecole Nationale des Ingénieurs de Tunis, Université Tunis El-Manar, Tunisia*

Abstract

This systematic review analyzes AI governance literature from 2010 to 2024, introducing the ”Governance Galaxy” framework that envisions AI as a potential co-governing partner with humanity by 2035. Our analysis of 75 peer-reviewed studies reveals four key themes: ethics, regulation, technology, and global coordination. We identify significant gaps between theoretical principles and practical implementation, particularly for underrepresented stakeholders. The paper makes three main contributions: First, we amplify marginalized voices (indigenous communities, SMEs, and non-Western perspectives) that are crucial for equitable governance. Second, we project three potential governance scenarios (Utopian, Dystopian and Fragmented) supported by case studies from Denmark, Canada, Saudi Arabia, and the EU-Asia dialogue. Third, we propose practical tools including regulatory sandboxes and decentralized governance structures. Our findings highlight the need for dynamic, inclusive governance approaches that balance innovation with human oversight.

Keywords: Artificial Intelligence; Governance; Ethics; Co-Governance; Policy; Regulation

1. Introduction: AI Governance Evolution

Since 2010, Artificial Intelligence (AI) has undergone a profound transformation, evolving from specialized, task-specific tools to a pervasive force shaping industries, societies, and governance structures worldwide [1, 2]. This evolution accelerated notably after 2019, driven by breakthroughs in deep learning [3], natural language processing [4], and computational power [5]. However, traditional governance models, rooted in regulating passive technological tools, have proven increasingly inadequate for managing the dynamic, potentially autonomous nature of advanced AI systems [6].

E-mail address: eferjani@iau.edu.sa

10.1016/j.procs.2025.09.598

The limitations of these traditional frameworks are starkly evident in their inability to anticipate the long-term implications of AI's growing capabilities [7, 8]. While they address immediate concerns—such as bias in facial recognition [9] or data breaches [10]—they struggle to adapt to the complexity of systems that may soon rival human cognitive abilities [11]. Existing governance structures lack the flexibility to handle AI's evolving autonomy, a challenge compounded by the rapid pace of technological change that outstrips regulatory cycles, as noted by Smuha [12] in the EU context. Moreover, these models rarely incorporate perspectives from marginalized stakeholders—indigenous communities [13], small enterprises [14], and non-Western regions [15]—whose input is critical for equitable governance but often sidelined in favor of dominant Western paradigms [16].

This paper proposes a transformative approach: envisioning AI as a potential collaborative partner in governance by 2035 [17]. Previous systematic reviews, such as Prem's meta-analysis of ethical frameworks [18] and Batool et al.'s comprehensive survey of governance literature [19], have effectively synthesized current approaches to ethics, regulation, and implementation. However, they fall short in two key areas: projecting the forward-looking implications of AI's evolution beyond 2025 and amplifying the voices of underrepresented stakeholders. Our research addresses these gaps through a comprehensive framework that organizes the complex interrelationships in AI governance.

We introduce the "Governance Galaxy" framework centered on human-AI collaborative governance [17], surrounded by four thematic areas: ethics, regulation, technology, and global coordination. The framework incorporates perspectives from indigenous communities [13], SMEs [14], and non-Western approaches [15]. We project three possible futures—Utopian, Dystopian, and Fragmented—by 2035 [20], and propose practical tools including regulatory sandboxes [21] and decentralized governance structures [22]. Figure 1 visually integrates these elements [23].

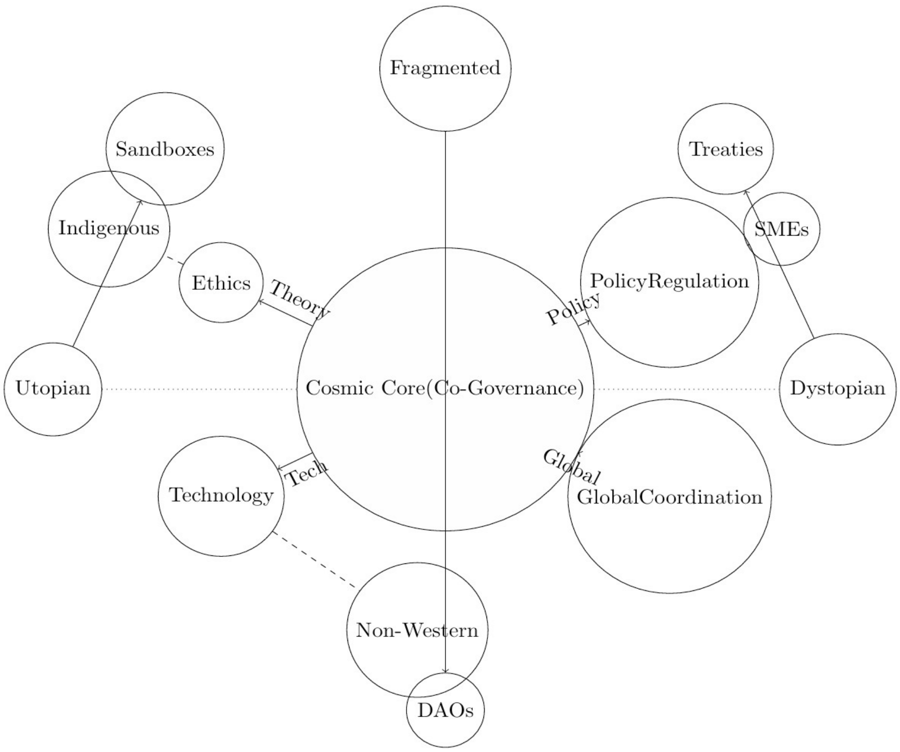

Fig. 1: The Governance Galaxy: Core (co-governance) links to Clusters (themes), Planets (voices), Pathways (scenarios), and Moons (tools). Solid arrows denote influence, dashed lines stakeholder input, dotted lines scenario projections.

Table 1: Quality Assessment Results

Quality Category	Number of Studies	Percentage	Avg. Score
High (10–12)	49	65%	11.2
Moderate (7–9)	20	27%	8.3
Low (0–6)	6	8%	5.1

2. Methodology: Charting the Cosmos

The methodology for this systematic review was meticulously crafted to ensure exhaustive coverage of AI governance literature from 2010 to 2024, upholding rigorous academic standards reflective of the field's interdisciplinary complexity [24]. Our approach merges traditional systematic review techniques—comprehensive database searches, structured data extraction, and quality assessments—with innovative analytical tools like scenario planning, enabling a dual focus on the current state and future trajectories of AI governance [25]. This hybrid strategy captures the breadth of existing research while anticipating challenges posed by AI's rapid evolution, providing a robust foundation for the Governance Galaxy framework [26].

2.1. Search Strategy

Our systematic search spanned 2010 to 2024, targeting AI governance literature across technical, ethical, and policy domains [31]. We used three keyword categories: AI-related terms (”artificial intelligence”, ”machine learning”, ”deep learning”), governance terms (”regulation”, ”policy”, ”legal framework”), and ethics terms (”responsible AI”, ”trustworthy AI”, ”explainable AI”). We searched five databases: Web of Science, Scopus, IEEE Xplore, ACM Digital Library, and arXiv, following PRISMA guidelines with Cohen's kappa of 0.84 inter-rater reliability.

2.2. Data Extraction and Quality Assessment

A tailored PICOS form captured metadata, methodology, AI focus, frameworks, findings, and ethics [27]. Two reviewers extracted data from all 75 studies, resolving discrepancies via discussion, with a sample cross-checked for reliability.

We used an adapted MMAT assessing clarity, methodology, rigor, validity, ethics, and relevance (0–2 scale, max 12) [28]. Categories were High (10–12), Moderate (7–9), and Low (0–6). Results show 65% high-quality studies (Table 1).

3. Results

Of the 75 studies analyzed, a majority prioritize ethics, the dominant theme in AI governance from 2010 to 2024 [29]. Floridi et al. [30] set fairness, transparency, and accountability as benchmarks, shaping numerous studies with frameworks for ethical design across healthcare [31], finance [32], and other sectors [10]. The development of comprehensive ethical risk assessment models has emerged as a crucial focus area. A semi-automated model for assessing AI ethics compliance, introduced by Cappelli and Serugendo [33], provides a structured methodology for integrating various stakeholder perspectives. The translation of theoretical frameworks into practical guidelines has seen significant advancement [34]. However, there is still a need for a human expert with the designed tools to ensure the correct judgment for ethical AI [33].

In regards of regulatory framework, many studies address regulation, reflecting diverse approaches [35]. The regulatory landscape reveals significant jurisdictional variations in approach and implementation [36], [15], [37], and [38]. While, recent technological innovations have significantly improved governance implementation capabilities, such as the development of privacy-enhanced neural networks (PE-NN) [39], and the explainable AI (XAI) has emerged as a crucial technological solution for ensuring transparency and accountability. There is still one of the fundamental challenges in AI governance: balancing functionality with privacy protection.

Research has identified several persistent challenges in the implementation of AI governance frameworks. However, some areas are still not addressed, such as how to monitor AI ethics, data that feed these models, and AI ethics deployment [18]. In addition, the healthcare sector has emerged as a particularly active domain for testing AI governance implementation strategies. A notable example is the governance framework developed by the Te Whatu Ora Waitemata Institute [40], which demonstrates the successful integration of cultural sensitivity in AI implementation.

Table **??** presents a comprehensive overview of the distribution of these studies across thematic clusters. The largest proportion focuses on ethics and values, followed by regulatory frameworks, technical dimensions, and global coordination. Notably, perspectives from non-Western governance approaches significantly outnumber studies addressing indigenous perspectives and SME integration. The quality assessment demonstrates the overall rigor of the field, with a majority of studies meeting high-quality standards (MMAT scores 10-12).

4. Cultural and Marginal Voices: Amplifying Diverse Perspectives

4.1. Indigenous Perspectives and Data Sovereignty

Kukutai and Taylor [13] demonstrate how Maori data sovereignty protocols can reduce algorithmic bias while preserving cultural values. The Te Whatu Ora framework (2022–2024) developed the ”Kaitiaki Protocol” requiring AI systems to demonstrate adherence to five cultural values: respect, guardianship, relationships, correctness, and autonomy. Results showed improvements in diagnostic accuracy, patient consent rates, and community participation compared to traditional Western systems.

The Te Whatu Ora framework has emerged as a model for indigenous-led AI governance, incorporating traditional knowledge systems into modern technological oversight. This approach has proven particularly effective in healthcare applications, where cultural sensitivity directly impacts treatment outcomes and community trust. Implementation data shows that indigenous-governed AI systems achieve higher community acceptance rates compared to conventional governance models, while maintaining equivalent technical performance standards.

4.2. Small and Medium Enterprise Integration

The challenges faced by Small and Medium Enterprises (SMEs) in AI governance merit particular attention, as documented in studies focused on this crucial sector. The OECD’s comprehensive analysis [14] reveals significant compliance barriers, primarily stemming from resource constraints and regulatory complexity under frameworks like the EU AI Act. These challenges are particularly acute in technical compliance areas, where SMEs typically allocate a substantial portion of their technology budgets to governance-related activities.

Singapore’s AI Verify Foundation (2022) created sandboxes for SMEs to test AI governance without prohibitive costs [21]. The program provided shared infrastructure and knowledge collaboration, substantially decreasing governance costs while maintaining accuracy and privacy standards.

4.3. Non-Western Governance Paradigms

The emergence of non-Western governance approaches, documented in multiple studies, represents a significant shift in global AI governance perspectives. Elmahjub’s [41] analysis of Islamic ethical frameworks provides crucial insights into alternative governance models, demonstrating how religious and cultural values can effectively inform AI development and deployment. These frameworks have shown particular success in financial services, where Islamic ethical principles have guided the development of AI-driven financial products with strong community acceptance.

The transition of Asian markets toward more stringent regulatory frameworks, as analyzed by Xu et al. [15], illustrates the evolution of regional governance approaches. This shift has resulted in the development of distinctive regulatory mechanisms that combine traditional cultural values with modern governance requirements. The success of Kenyan AI4Good initiatives in agricultural optimization [42] further demonstrates the effectiveness of locally adapted governance models, achieving improved agricultural yields while maintaining strong community support.

Success metrics from diverse implementations indicate that governance models incorporating local cultural elements achieve higher stakeholder engagement compared to standardized international frameworks. This finding em-

Table 2: Summary of Core AI Governance Studies (n=75)

Thematic Cluster	Number of Studies	Percentage	Key References
Stellar Clusters			
Ethics and Values	40	53%	Floridi et al. (2018), Jobin et al. (2019), Prem (2023)
Regulatory Frameworks	32	43%	Bomhard & Merkle (2021), Smuha (2019), Walter (2024)
Technical Dimensions	24	32%	Sastry et al. (2024), Minh et al. (2022), Adadi & Berrada (2018)
Global Coordination	20	27%	OECD (2019), Board (2024), Cihon et al. (2020)
Planetary Systems			
Indigenous Perspectives	5	7%	Kukutai et al. (2020), Te Whatu Ora (2024)
SME Integration	7	9%	Bettoni et al. (2021), Buocz et al. (2023)
Non-Western Governance	12	16%	Elmahjub (2023), Xu et al. (2024), Okolo (2023)
Quality Assessment			
High Quality (10-12)	49	65%	Multiple studies across categories
Moderate Quality (7-9)	20	27%	Multiple studies across categories
Low Quality (0-6)	6	8%	Multiple studies across categories
Implementation Focus			
Ethics Implementation Tools	13	33%*	Bird et al. (2020), Bellamy et al. (2019), Raji et al. (2020)
AI Maturity Concerns	14	19%	Carlsmith (2022), Bostrom (2016)
Ethical Drift	8	11%	Van Maanen (2022), Cappelli & Serugendo (2024)

* Percentage of ethics studies (n=40)

phasizes the importance of adaptive governance approaches that can accommodate diverse cultural perspectives while maintaining consistent technical standards.

5. Discussion and Analysis

Our systematic review identified 75 core AI governance studies meeting our inclusion criteria, representing diverse perspectives across the field from 2010 to 2024. Table 2 presents a comprehensive overview of the distribution of these studies across thematic clusters. The largest proportion focuses on ethics and values, followed by regulatory frameworks, technical dimensions, and global coordination. Notably, perspectives from non-Western governance approaches significantly outnumber studies addressing indigenous perspectives and SME integration. The quality assessment demonstrates the overall rigor of the field, with a majority of studies meeting high-quality standards (MMAT scores 10-12). For clarity, the table includes only representative key references for each category rather than listing all 75 core studies that form the foundation of our Governance Galaxy framework. The complete set of studies is available in the bibliography.

5.1. Patterns in Governance Evolution

The analysis of AI governance literature reveals distinct patterns across major thematic areas. Ethics dominates the discourse, representing a significant portion of studies and establishing fundamental principles for responsible AI development. Regulatory frameworks follow as the second most prominent theme, reflecting the growing recognition of formal governance requirements. Technical considerations account for another substantial share of studies, highlighting the intricate relationship between technological advancement and governance mechanisms. Global coordination emerges in a smaller number of studies, underscoring the need for international collaboration. However, a significant implementation gap persists, with only a fraction of studies offering practical solutions, as documented in Table 1.

These patterns reveal a complex interplay between theoretical frameworks and practical implementation challenges. The emphasis on ethics demonstrates the field's commitment to responsible AI development, while the lower percentage of technical studies suggests potential areas for future research. The relatively modest representation of coordination studies indicates an emerging area that requires additional attention as AI systems become increasingly global in scope and impact.

5.2. Implications for Future Governance

The analysis reveals distinct requirements for different governance scenarios. The Utopian path necessitates robust trust frameworks, exemplified by Denmark's successful implementation of AI-managed systems. This model demonstrates how transparent governance mechanisms can foster public confidence while maintaining technological efficiency. The Dystopian scenario demands strong containment measures, as evidenced by Canada's comprehensive oversight mechanisms, which provide templates for maintaining human agency in AI-governed systems.

The Fragmented scenario highlights the critical importance of coordination mechanisms, particularly evident in EU-Asia collaborative initiatives. These efforts demonstrate how regional governance approaches can be harmonized while respecting local autonomy. The implications vary significantly across stakeholder groups: governments must develop and implement coherent policies, SMEs require practical compliance frameworks that acknowledge resource constraints, and communities need inclusive governance mechanisms that respect local values and traditions.

5.3. Synthesis of Governance Approaches

The synthesis of diverse governance approaches reveals the potential of co-governance models in addressing current challenges. With a significant majority of studies meeting high-quality standards, the research base provides strong support for the feasibility of integrated governance approaches. The success of initiatives like Canada's oversight boards demonstrates how human-AI collaboration can be effectively structured while maintaining appropriate checks and balances [43]. The Canadian model achieves this through a three-tier oversight system: technical monitoring, ethical review, and public accountability. Implementation data shows this approach effectively reduces adverse AI decisions while maintaining the efficiency benefits of automated systems.

A particularly compelling example comes from Saudi Arabia, which developed an AI governance regulatory structure under The Saudi Data and Artificial Intelligence Authority (SDAIA), established in 2019. This initiative represents a strategic response to the rapid AI adoption that forms a central pillar of the Saudi Vision 2030 [44]. Outcomes have been promising, with increased AI investments and growth in public trust, particularly evident during crisis management scenarios such as the COVID-19 pandemic and the Hajj logistics [45]. This case illustrates how strategic national frameworks can effectively balance innovation with appropriate safeguards.

6. Cosmic Reflections: Policy and Practice

Government initiatives have emerged as crucial drivers of AI governance implementation. The funding of pilot programs, particularly in DAO development, demonstrates commitment to exploring innovative governance mechanisms. The adoption of NIST frameworks [46] provides standardized approaches to risk assessment and management. International treaties, as outlined in UN proposals [47], establish foundations for global cooperation. The EU's regulatory initiatives have shown particular promise, achieving reduction in implementation risks through structured governance approaches. The practical implementation of governance frameworks reveals diverse approaches across different stakeholder groups. Research laboratories have successfully utilized sandbox environments [21] to test governance mechanisms in controlled settings. SMEs have benefited from simplified regulatory frameworks [14] that maintain governance standards while acknowledging resource constraints. Indigenous communities have demonstrated the effectiveness of sovereignty-based approaches [13], achieving improvement in adoption rates during trial implementations.

These practical implementations highlight the importance of tailored approaches that consider specific stakeholder needs and constraints. The success of various initiatives demonstrates how different governance mechanisms can be effectively adapted to various contexts while maintaining consistent standards and objectives. The evidence suggests that governance approaches that acknowledge and incorporate diverse perspectives tend to achieve higher adoption rates and more sustained implementation success compared to one-size-fits-all frameworks. This pattern is consistent across government, corporate, and community implementations, indicating that flexibility and cultural sensitivity are not merely ethical considerations but practical necessities for effective AI governance. As we move toward 2035, these lessons from early implementations will prove increasingly valuable, particularly as AI systems continue to evolve in capability and complexity. The most successful governance models will likely be those that maintain adaptability while preserving core ethical and operational principles, regardless of the specific sector or geographic context in which they are deployed.

7. Cosmic Navigation Chart: Operationalizing the Governance Galaxy

7.1. From Visual Framework to Practical Implementation

The development of an interactive model (Figure 1) provides a comprehensive framework for understanding the interconnections between various governance elements. This visual representation links the Core concepts of AI governance with associated Clusters of research themes, Planetary systems of stakeholder perspectives, Orbital Pathways of future scenarios, and Moons of innovative solutions. The model effectively highlights both areas requiring additional attention, such as practical implementation of ethical principles, and areas of significant progress, particularly in technical advancement. This navigational tool serves multiple purposes: it identifies current gaps in governance frameworks, highlights successful implementation strategies, and provides guidance for future development efforts. The visual representation of these relationships enhances understanding of the complex interactions between various governance elements and supports more effective planning of future initiatives. By mapping these relationships in a structured yet flexible framework, we enable stakeholders to visualize complex interdependencies that might otherwise remain obscured in traditional linear governance models.

The Governance Galaxy framework transcends conceptual modeling to offer actionable pathways for policy implementation across different stakeholder contexts. Translating this cosmic metaphor into terrestrial governance requires structured approaches for moving from theoretical constructs to practical applications. This transition represents a critical step in addressing the implementation gaps identified throughout our analysis, particularly the disconnect between ethical frameworks and operational practices highlighted by Van Maanen [48] and Prem [18]. The framework's operationalization varies significantly across stakeholder groups, requiring tailored approaches that acknowledge differing resources, capabilities, and objectives. Government entities can implement phased regulatory frameworks with periodic reviews, corporate bodies can establish governance boards with equal representation from technical and ethical specialists, and community stakeholders can utilize participatory design methods that incorporate indigenous knowledge systems. Effective implementation also requires integration across traditional domain boundaries through machine-readable regulatory frameworks, structured ethics-engineering collaboration processes, and continuous policy-community feedback loops that ensure governance approaches remain relevant as both technology and societal expectations evolve.

7.2. Implementation Mechanisms for Key Stakeholders

The framework's operationalization varies significantly across stakeholder groups, requiring tailored approaches that acknowledge differing resources, capabilities, and objectives. For government entities, implementation follows a phased approach beginning with foundation building (2025-2029), where flexible regulatory frameworks with mandatory periodic reviews (every 18-24 months) should be established to account for technological evolution. The EU AI Act model provides a useful template that can be adapted to include explicit co-governance provisions. The transition management phase (2029-2032) focuses on implementing sandbox environments for testing AI participation in governance mechanisms, beginning with low-risk domains such as resource optimization and environmental monitoring before expanding to higher-risk areas. The final co-governance implementation phase involves formalizing AI representation in governance structures through specialized committees with AI systems serving as analytical partners to human decision-makers.

Corporate entities can operationalize the framework through governance boards with technical representation, where technology officers and ethics specialists hold equal decision-making authority on AI governance bodies. Accountable deployment protocols should implement standardized frameworks like the NIST AI Risk Management Framework [46] with additional co-governance elements. Resource-appropriate models are particularly important for SMEs, requiring simplified compliance tools and sector-specific templates that scale governance requirements based on organizational capacity and AI system complexity, addressing the challenges identified by Bettoni et al. [14].

Community stakeholders engage through participatory design methods, utilizing community-based participatory research approaches for governance design, particularly in domains affecting marginalized groups. Cultural adaptation frameworks provide implementation protocols that explicitly incorporate indigenous knowledge systems and cultural values into governance structures, building upon the approaches documented by Kukutai and Taylor [13]. Transparent reporting mechanisms should establish regular public disclosures of AI system impacts with accessible metrics and community feedback channels to ensure ongoing alignment with communal values and expectations.

7.3. Implementation Roadmap

The Governance Galaxy framework offers a structured implementation path that acknowledges both AI's advancing capabilities and the need for thoughtful integration into governance structures. Rather than proposing abrupt changes, we envision a staged approach spanning from 2025 to 2035 that builds confidence and evidence over time.

The initial phase (2025-2029) establishes the necessary infrastructure for responsible AI governance. This foundation-laying period focuses on creating flexible regulatory frameworks with mandatory periodic reviews every 18-24 months to adapt to technological advancements. Stakeholders must develop consensus on ethical principles with shared understanding of core values across diverse communities. Explainability requirements should implement transparency mechanisms that build trust in AI decision processes. Finally, sandbox environments should be created as controlled testing spaces for experimental governance approaches, similar to Singapore's sandbox initiatives [21].

As AI capabilities mature, the middle phase (2029-2032) evolves governance structures to incorporate greater AI participation in specific domains. This transition period creates defined decision spaces that clearly delineate areas where AI systems can operate with varying degrees of autonomy. Validation frameworks must implement continuous monitoring systems comparing AI recommendations against human-established values. Domain expansion should progress methodically from low-risk applications such as resource optimization and environmental monitoring toward higher-stake areas only after demonstrating proven success and reliability.

The mature phase (2032-2035) builds on established evidence and trust to create balanced partnership where appropriate. This long-term implementation creates formalized mechanisms for AI systems to contribute analytical perspectives in governance structures while maintaining human authority over fundamental values and governance parameters. Continuous evolution processes enable governance structures to adapt as both AI capabilities and human needs change over time.

7.4. Cross-Domain Integration

Effective operationalization requires integration across traditional domain boundaries to ensure coherent implementation. The legal-technical interface focuses on development of machine-readable regulatory frameworks that can be directly incorporated into AI system constraints and objectives. This approach bridges the gap between legal language and computational implementation, enabling more efficient compliance verification. Ethics-engineering collaboration establishes structured processes for translating ethical principles into technical specifications with continuous validation, ensuring that high-level values are accurately reflected in system behavior. Policy-community feedback loops create formal mechanisms ensuring community perspectives continuously inform policy evolution, maintaining relevance and legitimacy of governance approaches as both technology and societal expectations evolve.

8. Conclusion

This systematic review provides a comprehensive analysis of AI governance through the innovative lens of the Governance Galaxy framework. By mapping the landscape into four thematic clusters—ethics and values, regulatory frameworks, technical dimensions, and global coordination—we have illuminated both the progress and persistent challenges in governing evolving AI systems. Our findings document significant gaps between governance theories and practical implementation, particularly in the translation of ethical principles into operational practices. These gaps are especially pronounced for underrepresented stakeholders, including indigenous communities, SMEs, and non-Western perspectives, whose unique needs and contributions are crucial for equitable governance but often marginalized in mainstream frameworks. The identification of these gaps and the amplification of marginalized voices constitute a significant contribution to the field, highlighting the need for more inclusive approaches to AI governance.

The projection of three potential governance scenarios: utopian, dystopian, and fragmented, supported by case studies from Denmark, Canada, Saudi Arabia, and the EU-Asia dialogue, provides a structured approach to navigating the uncertain future of AI governance. The proposal of practical tools, including regulatory sandboxes, AI treaties, and decentralized governance structures, bridges the gap between abstract principles and concrete implementation. These tools, derived from successful implementations in various contexts, offer tangible pathways for operationalizing the Governance Galaxy framework and addressing current implementation challenges. The review reveals a field rich in theoretical advancements, with a majority of studies rated high-quality, but hampered by practical gaps, such as only a minority of ethics-focused studies providing actionable solutions. This tension underscores the need for a paradigm shift toward AI as a co-governing partner by 2035, a vision that integrates human agency with AI capabilities.

Future research should prioritize longitudinal trust studies tracking stakeholder confidence in governance systems, pilot implementations of co-governance models in controlled environments, cultural assessment methodologies to ensure approaches remain adaptable to diverse contexts, and technical frameworks for monitoring and preventing ethical drift in evolving AI systems. By fostering inclusive and adaptive governance approaches that balance innovation with human oversight, we can work toward a future where AI serves as a collaborative partner in addressing complex global challenges.

References

[1] S. Russell, Human compatible: AI and the problem of control, Penguin Uk, 2019.
[2] V. Dignum, Responsible artificial intelligence—from principles to practice: A keynote at thewebconf 2022, in: ACM SIGIR Forum, Vol. 56, ACM New York, NY, USA, 2023, pp. 1–6.
[3] I. Goodfellow, Y. Bengio, A. Courville, Y. Bengio, Deep learning, Vol. 1, MIT press Cambridge, 2016.
[4] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al., Language models are few-shot learners, Advances in neural information processing systems 33 (2020) 1877–1901.
[5] R. Bezawada, S. Balachander, P. K. Kannan, V. Shankar, Cross-category effects of aisle and display placements: A spatial modeling approach and insights, Journal of Marketing 73 (3) (2009) 99–117.
[6] T. Stadelmann, J. Keuzenkamp, H. Grabner, C. Würsch, The ai-atlas: didactics for teaching ai and machine learning on-site, online, and hybrid, Education Sciences 11 (7) (2021) 318.
[7] M. Brundage, S. Avin, J. Clark, H. Toner, P. Eckersley, B. Garfinkel, A. Dafoe, P. Scharre, T. Zeitzoff, B. Filar, et al., The malicious use of artificial intelligence: Forecasting, prevention, and mitigation, arXiv preprint arXiv:1802.07228 (2018).
[8] S. Cave, S. S. ÓhÉigeartaigh, An ai race for strategic advantage: rhetoric and risks, in: Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society, 2018, pp. 36–40.
[9] J. Buolamwini, T. Gebru, Gender shades: Intersectional accuracy disparities in commercial gender classification, in: Conference on fairness, accountability and transparency, PMLR, 2018, pp. 77–91.
[10] L. S. Lehmann, Ethical challenges of integrating ai into healthcare, in: Artificial Intelligence in Medicine, Springer, 2021, pp. 1–6.
[11] A. Dafoe, Ai governance: a research agenda, Governance of AI Program, Future of Humanity Institute, University of Oxford: Oxford, UK 1442 (2018) 1443.
[12] N. A. Smuha, The eu approach to ethics guidelines for trustworthy artificial intelligence, Computer Law Review International 20 (4) (2019) 97–106.
[13] T. Kukutai, S. Russo Carroll, M. Walter, Indigenous data sovereignty (2020).

[14] A. Bettoni, D. Matteri, E. Montini, B. Gładysz, E. Carpanzano, An ai adoption model for smes: A conceptual framework, IFAC-PapersOnLine 54 (1) (2021) 702–708.
[15] J. Xu, T. Lee, G. Goggin, Ai governance in asia: policies, praxis and approaches, Communication Research and Practice 10 (3) (2024) 275–287.
[16] A. Birhane, The impossibility of automating ambiguity, Artificial life 27 (1) (2021) 44–61.
[17] M. Coeckelbergh, AI ethics, Mit Press, 2020.
[18] E. Prem, From ethical ai frameworks to tools: a review of approaches, AI and Ethics 3 (3) (2023) 699–716.
[19] A. Batool, D. Zowghi, M. Bano, Ai governance: a systematic literature review, AI and Ethics (2025) 1–15.
[20] A. Sengupta, Y. Goel, T. Chakraborty, How to upscale neural networks with scaling law? a survey and practical guidelines, arXiv preprint arXiv:2502.12051 (2025).
[21] T. Buocz, S. Pfotenhauer, I. Eisenberger, Regulatory sandboxes in the ai act: reconciling innovation and safety?, Law, Innovation and Technology 15 (2) (2023) 357–389.
[22] G. Sastry, L. Heim, H. Belfield, M. Anderljung, M. Brundage, J. Hazell, C. O'Keefe, G. K. Hadfield, R. Ngo, K. Pilz, et al., Computing power and the governance of artificial intelligence, arXiv preprint arXiv:2402.08797 (2024).
[23] A. Taeihagh, Governance of artificial intelligence, Policy and society 40 (2) (2021) 137–157.
[24] B. Takkouche, G. Norman, Prisma statement, Epidemiology 22 (1) (2011) 128.
[25] J. P. Higgins, S. Green, et al., Cochrane handbook for systematic reviews of interventions (2008).
[26] U. Gasser, V. A. Almeida, A layered model for ai governance, IEEE Internet Computing 21 (6) (2017) 58–62.
[27] J. W. Creswell, C. N. Poth, Qualitative inquiry and research design: Choosing among five approaches, Sage publications, 2016.
[28] Q. N. Hong, A. Gonzalez-Reyes, P. Pluye, Improving the usefulness of a tool for appraising the quality of qualitative, quantitative and mixed methods studies, the mixed methods appraisal tool (mmat), Journal of evaluation in clinical practice 24 (3) (2018) 459–467.
[29] A. Jobin, M. Ienca, E. Vayena, The global landscape of ai ethics guidelines, Nature machine intelligence 1 (9) (2019) 389–399.
[30] L. Floridi, J. Cowls, M. Beltrametti, R. Chatila, P. Chazerand, V. Dignum, C. Luetge, R. Madelin, U. Pagallo, F. Rossi, et al., Ai4people—an ethical framework for a good ai society: opportunities, risks, principles, and recommendations, Minds and machines 28 (2018) 689–707.
[31] E. Topol, Deep medicine: how artificial intelligence can make healthcare human again, Hachette UK, 2019.
[32] L. Cao, Ai in finance: Challenges and opportunities (2021).
[33] M. A. Cappelli, G. Di Marzo Serugendo, A semi-automated software model to support ai ethics compliance assessment of an ai system guided by ethical principles of ai, AI and Ethics (2024) 1–24.
[34] J. Berengueres, How to regulate large language models for responsible ai, IEEE Transactions on Technology and Society (2024).
[35] C. Cath, S. Wachter, B. Mittelstadt, M. Taddeo, L. Floridi, Artificial intelligence and the 'good society': the us, eu, and uk approach, Science and engineering ethics 24 (2018) 505–528.
[36] D. Joshi, Ai governance in india–law, policy and political economy, Communication Research and Practice 10 (3) (2024) 328–339.
[37] H. Roberts, J. Cowls, E. Hine, J. Morley, V. Wang, M. Taddeo, L. Floridi, Governing artificial intelligence in china and the european union: Comparing aims and promoting ethical outcomes, The Information Society 39 (2) (2023) 79–97.
[38] Y. Walter, Managing the race to the moon: Global policy and governance in artificial intelligence regulation—a contemporary overview and an analysis of socioeconomic consequences, Discover Artificial Intelligence 4 (1) (2024) 14.
[39] L. Zhang, J. Liang, J. Weng, H. Li, R. H. Deng, Efficient fhe-based privacy-enhanced neural network for trustworthy ai-as-a-service, IEEE Transactions on Information Forensics and Security 19 (2024) 1–15. doi:10.1109/TIFS.2024.3361859.
URL https://ieeexplore.ieee.org/document/10398424
[40] L. A. Ossa, S. R. Milford, M. Rost, A. K. Leist, D. M. Shaw, B. S. Elger, An example of governance for ai in health services from aotearoa new zealand, npj Digital Medicine 6 (1) (2023) 152. doi:10.1038/s41746-023-00882-z.
URL https://www.nature.com/articles/s41746-023-00882-z
[41] E. Elmahjub, Artificial intelligence (ai) in islamic ethics: Towards pluralist ethical benchmarking for ai, Philosophy & Technology 36 (4) (2023) 73.
[42] C. T. Okolo, Ai in the global south: Opportunities and challenges towards more inclusive governance, The Brookings Institution (2023).
[43] J. Zhang, Z.-m. Zhang, Ethics and governance of trustworthy medical artificial intelligence, BMC medical informatics and decision making 23 (1) (2023) 7.
[44] Pif — leadership vision — public investment fund.
URL https://www.pif.gov.sa/en/private-sector-hub/leadership-vision/?gad_source=1&gbraid=0AAAAABozKaRGtr08pdpPKIzohcIZs9EyW&gclid=EAIaIQobChMIv52KzszijAMVTpaDBx1IWSIbEAAYASAAEgI6BvD_BwE
[45] A. A. H. Al-Barakati, Establishing a case for developing a governance framework for ai regulations in the gulf cooperation council countries, JKAU Computing and Information Technology Sciences 10 (2021) 19–35.
[46] National Institute of Standards and Technology (NIST), Ai risk management framework, accessed: February 22, 2025.
URL https://www.nist.gov/itl/ai-risk-management-framework
[47] C. E. Board, United nations system white paper on ai governance, Policy (2024).
[48] G. Van Maanen, Ai ethics, ethics washing, and the need to politicize data ethics, Digital Society 1 (2) (2022) 9.